

ME1103

Principles of Mechanics and Materials

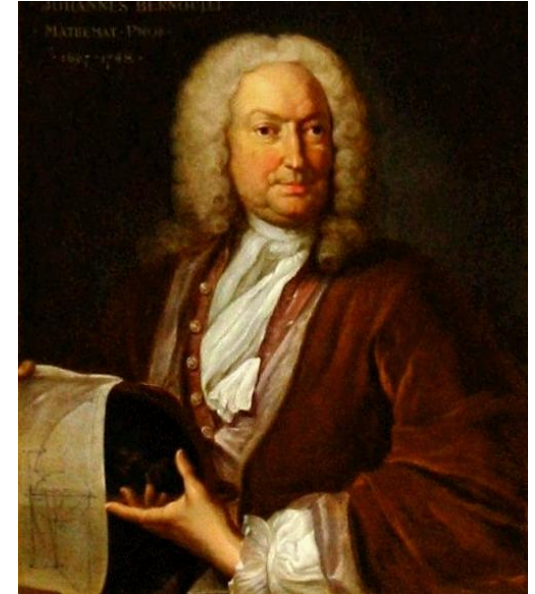
PRINCIPLE OF VIRTUAL WORK

Principle of Virtual Work

- So far, the statics analysis is based on Newton's laws, which is a vector analysis. The equations of equilibrium are given by:

$$\sum \vec{F}_i = 0, \text{ and } \sum \vec{M}_i = 0$$

- Here, we will introduce another approach called the Principle of Virtual Work, which was first formulated by Jean Bernoulli in mechanics study.



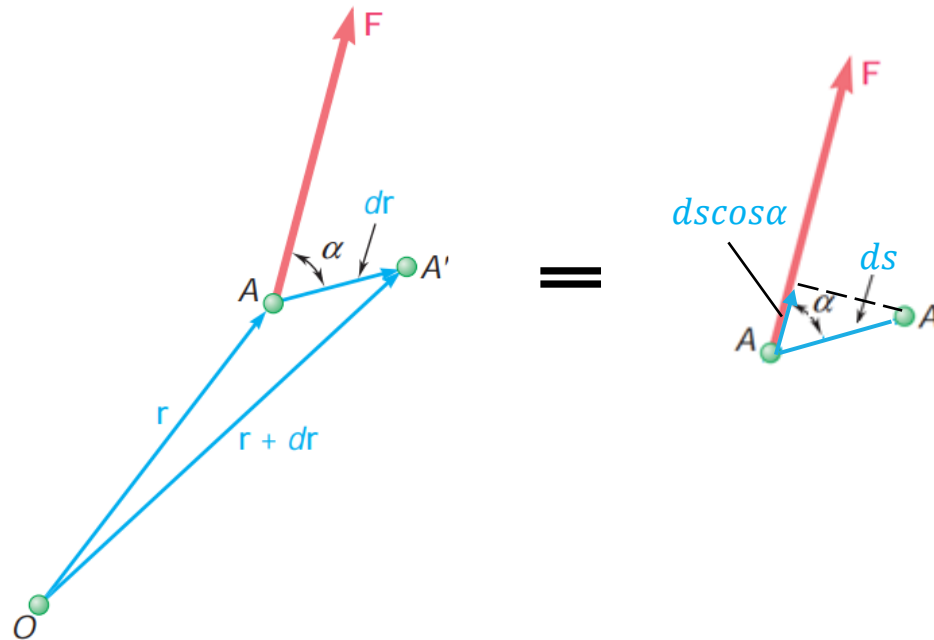
Principle of virtual work states that if a particle or rigid body, or, more generally, a system of connected rigid bodies, which is in equilibrium under various external forces, is given an arbitrary displacement from that position of equilibrium, the total work done by the external forces during the displacement is zero.

Work of a Force

- The work of a force $\vec{F}$ corresponding to an infinitesimal displacement $d\vec{r}$ is defined as

$$dU = \vec{F} \cdot d\vec{r} = F ds \cos \alpha$$

where F and ds are the magnitudes of the force and displacement, respectively, and α is the angle between the two vectors.



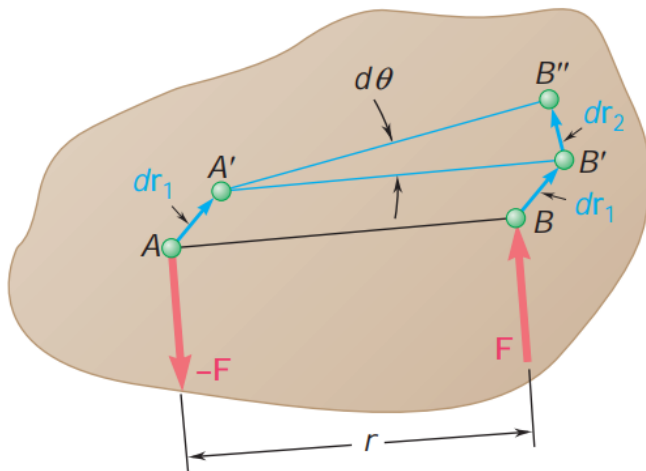
Work of a Couple

- Consider a couple that is applied on a rigid body, which is defined by

$$\vec{M} = \vec{r} \times \vec{F}$$

whose magnitude for 2D planar problem is given by $M = Fr$.

- Suppose it undergoes a general infinitesimal displacement that comprises a linear displacement of $d\vec{r}_1$ and rotation of $d\theta$.



- The work of the couple is then given by

$$dU = -\vec{F} \cdot d\vec{r}_1 + \vec{F} \cdot (d\vec{r}_1 + d\vec{r}_2) = \vec{F} \cdot d\vec{r}_2$$

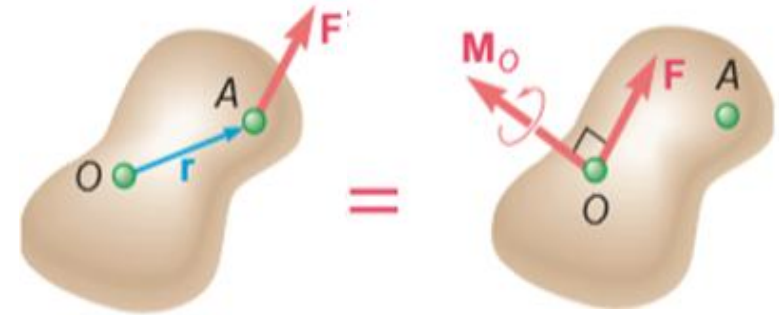
where $d\vec{r}_2$ is tangent to $\vec{r}$ and hence parallel to the force. The magnitude of $d\vec{r}_2$ is the arc length $B'B''$, which is given by $ds_2 = r d\theta$ for small angle $d\theta$.

- With this, the work done becomes

$$dU = F ds_2 = F(r d\theta) = M d\theta$$

Work of Forces under General Displacement

- Recall that the effect of a force applied on a rigid body at point A can always be represented as the effect of a force-couple pair about any other point O .



- Hence, the work of the force that undergoes a general infinitesimal displacement of $d\vec{r}_O$ (translation at O) and $d\theta$ (identical at all points) would be given by

$$dU = \vec{F} \cdot d\vec{r}_A = \vec{F} \cdot d\vec{r}_O + M d\theta$$

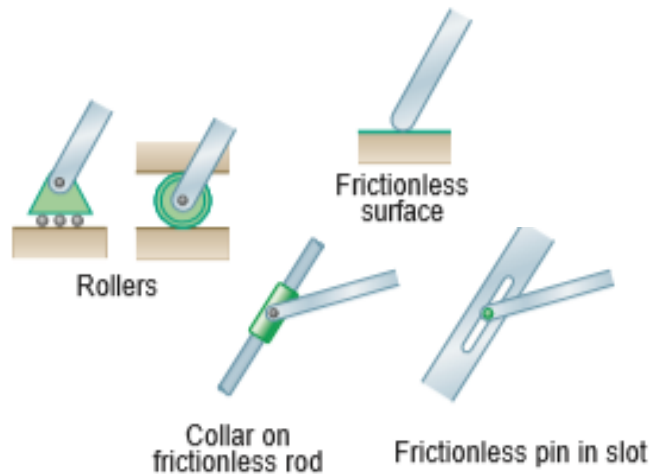
- Suppose there are multiple forces applying on the rigid body. Then the work done by all the forces becomes

$$dU = \sum \vec{F}_i \cdot d\vec{r}_i = \sum (\vec{F}_i \cdot d\vec{r}_O + M_i d\theta) = \left(\sum \vec{F}_i \right) \cdot d\vec{r}_O + \left(\sum M_i \right) d\theta$$

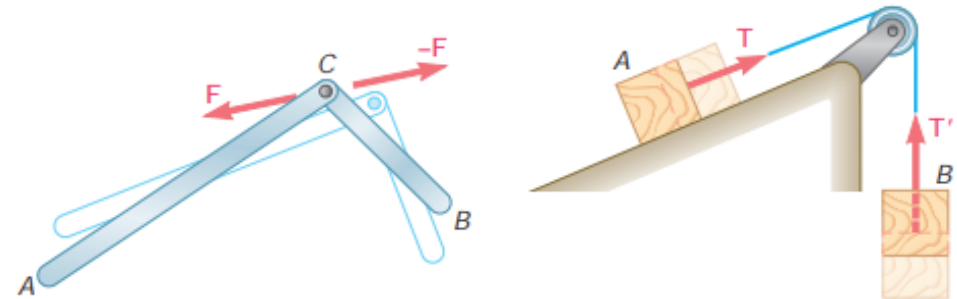
Forces that Do NO Work

➤ Some common situations in statics where the forces do no works:

- Force applied to fixed supports, i.e. $ds = 0$.



- The direction of the force is perpendicular to the displacement.



- Internal forces are action-reaction pair, and hence they always do equal and opposite work.

Principle of Virtual Work

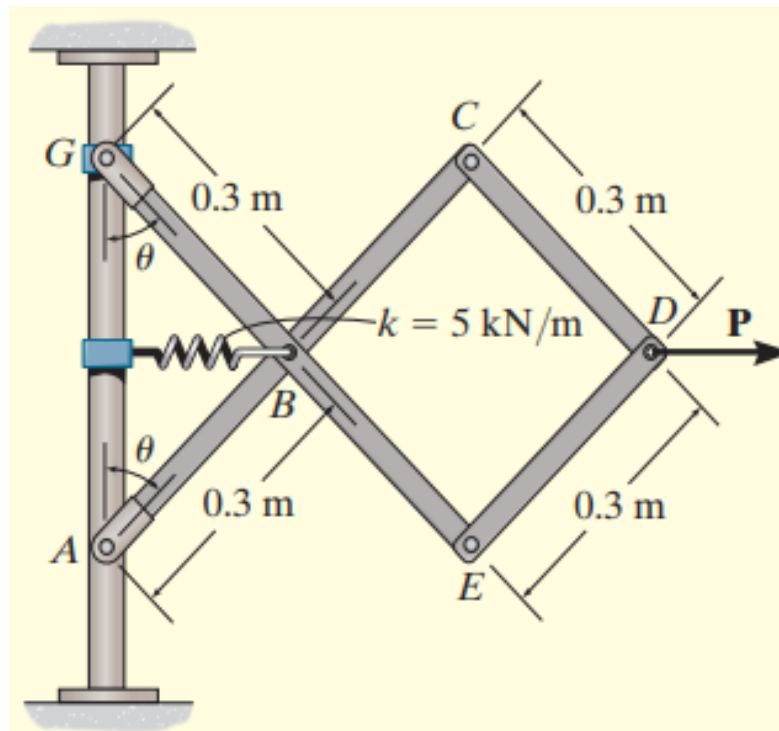
- Suppose the displacement did not actually take place. Instead, it is an imaginary one that we perceive to occur after the rigid body attains its static equilibrium.
- We called this imaginary displacement virtual displacement, which is denoted by $\delta \vec{r}$.
- The corresponding work of the external forces are also virtual, hence referred to as virtual work, which is expressed as

$$dU = \left(\sum \vec{F}_i \right) \cdot \delta \vec{r}_O + \left(\sum M_i \right) \delta \theta$$

- Finally, for a rigid body that is in static equilibrium, i.e. $\sum \vec{F}_i = \sum M_i = 0$, then the virtual work must be equal to zero, i.e. $dU = 0$.
- This concept can be extended to a machine, since all the components must also satisfy this condition ($dU = 0$), if the assembly is at static equilibrium.

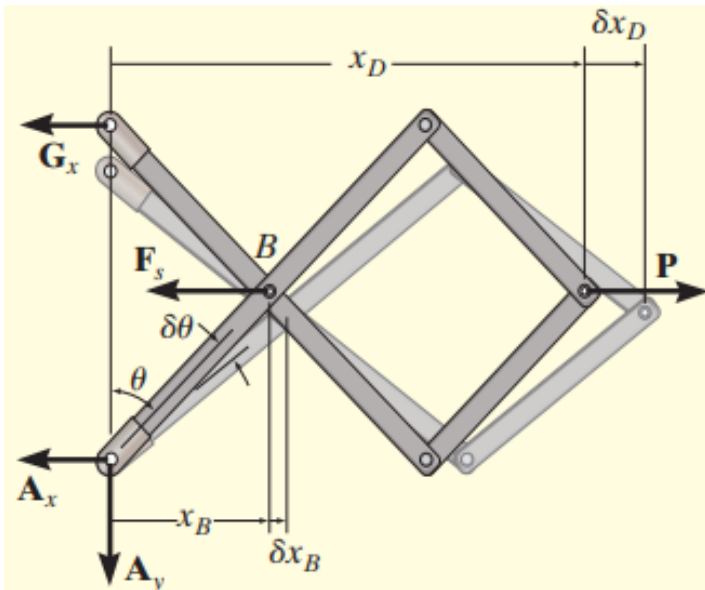
Example 5.1

The figure shows a scissor linkage which is fixed at A by a pin and connected to a collar at G that can slide freely along the vertical rod. Determine the force P needed to maintain static equilibrium of the mechanism when $\theta = 60^\circ$. Note that the spring is unstretched when $\theta = 30^\circ$.



Example 5.1...

Consider the FBD of the mechanism which undergoes a virtual displacement $\delta\theta$.



Based on this virtual displacement, only the spring force F_s and P do work.

At the static equilibrium configuration, the spring force is given by

$$F_s = k\Delta x$$

$$= 5000(0.3\sin 60^\circ - 0.3\sin 30^\circ) = 549 \text{ N}$$

With reference to the fixed point A , and in terms of the displacement coordinate θ , the coordinates for B and D are:

$$x_B = 0.3\sin\theta$$

$$x_D = 3x_B = 0.9\sin\theta$$

Example 5.1...

Differentiating these expressions, we have the virtual displacements of B and D

$$\delta x_B = 0.3 \cos \theta \cdot \delta \theta$$

$$\delta x_D = 0.9 \cos \theta \cdot \delta \theta$$

Finally, by the principle of virtual work, where the total virtual work done by the external forces must be equal to zero, we have

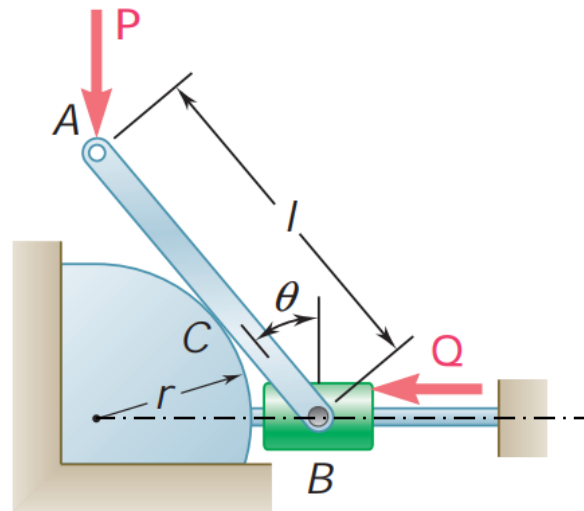
$$-F_s \cdot \delta x_B + P \cdot \delta x_D = 0$$

$$\Rightarrow -549(0.3 \cos \theta \cdot \delta \theta) + P(0.9 \cos \theta \cdot \delta \theta) = 0$$

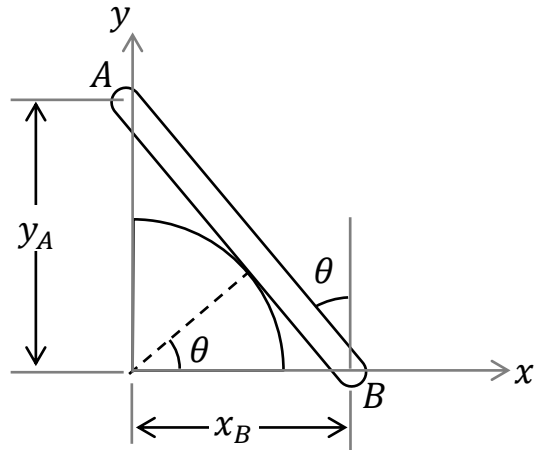
$$\Rightarrow P = \mathbf{183 \text{ N}}$$

Example 5.2

A slender rod of length l is attached to a collar at B and rests on a portion of a circular cylinder of radius r . Neglecting the effect of friction, determine the value of θ corresponding to the equilibrium position of the mechanism when $l = 200 \text{ mm}$, $r = 60 \text{ mm}$, $P = 40 \text{ N}$ and $Q = 80 \text{ N}$.



Example 5.2...



Based on the geometry, we can derive the following relations

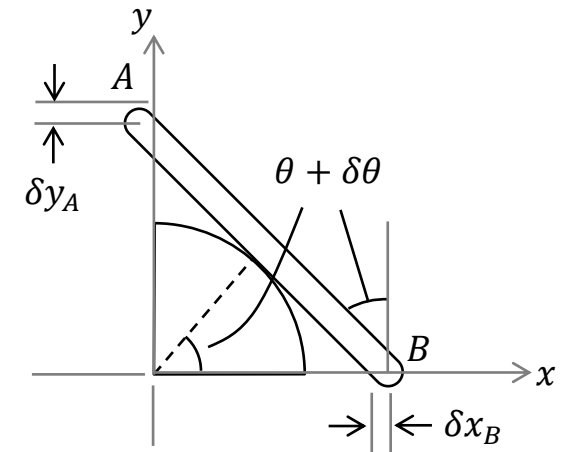
$$y_A = l \cos \theta$$

$$\cos \theta = \frac{r}{x_B} \Rightarrow x_B = \frac{r}{\cos \theta}$$

With these expression, an infinitesimal change in the angular position $\delta \theta$ will give infinitesimal displacement at A and B as

$$\delta y_A = \frac{d}{d\theta} (l \cos \theta) \delta \theta = -l \sin \theta \delta \theta$$

$$\delta x_B = \frac{d}{dx} \left(\frac{r}{\cos \theta} \right) \delta \theta = \left(\frac{r \sin \theta}{\cos^2 \theta} \right) \delta \theta$$



Example 5.2...

Next, by the principle of virtual work, we have

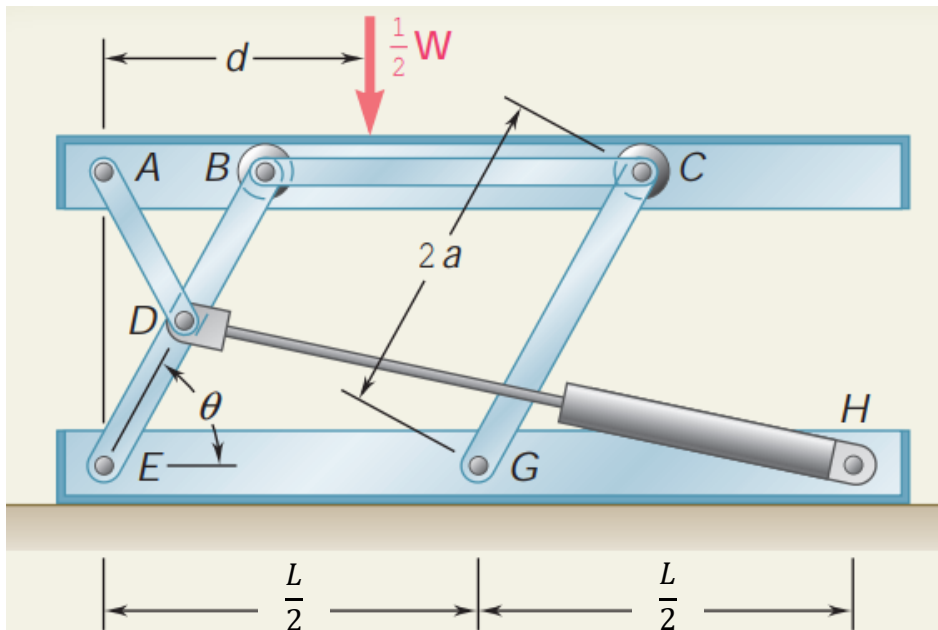
$$\begin{aligned}\delta U &= P\delta y_A + Q\delta x_B = 0 \\ \Rightarrow P(-l\sin\theta\delta\theta) + Q\left(\frac{r\sin\theta}{\cos^2\theta}\right)\delta\theta &= 0 \\ \Rightarrow \cos^2\theta &= \frac{Qr}{Pl}\end{aligned}$$

Finally, substituting the numeric, gives

$$\cos^2\theta = \frac{(80)(60)}{(40)(200)} \Rightarrow \theta = 39.23^\circ$$

Example 5.3 (Redo Example 4.7)

A hydraulic-lift table is used to raise a 1000 *kg* crate. It consists of a platform and two identical linkages on which hydraulic cylinders exert equal forces. Members *EDB* and *CG* are each of length $2a$, and member *AD* is pinned to the midpoint of *EDB*.

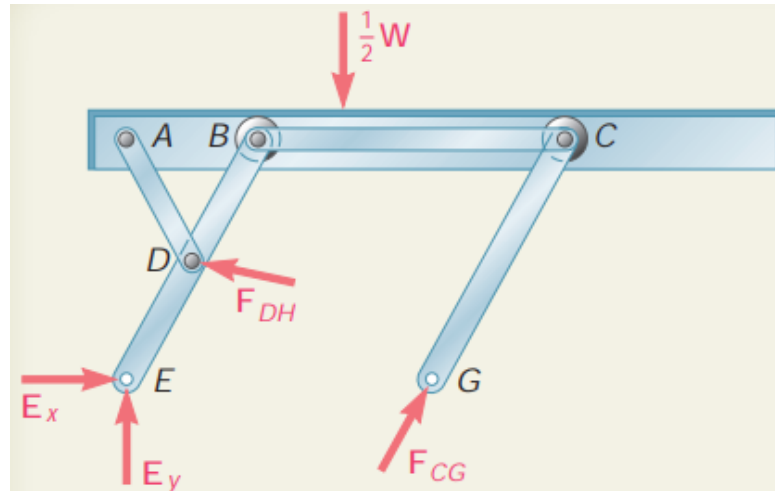


If the crate is placed on the table, so that half of its weight is supported by the system shown, determine the force exerted by each cylinder for:

$$\theta = 60^\circ, a = 0.70 \text{ m}, \text{ and } L = 3.20 \text{ m}.$$

Example 5.3...

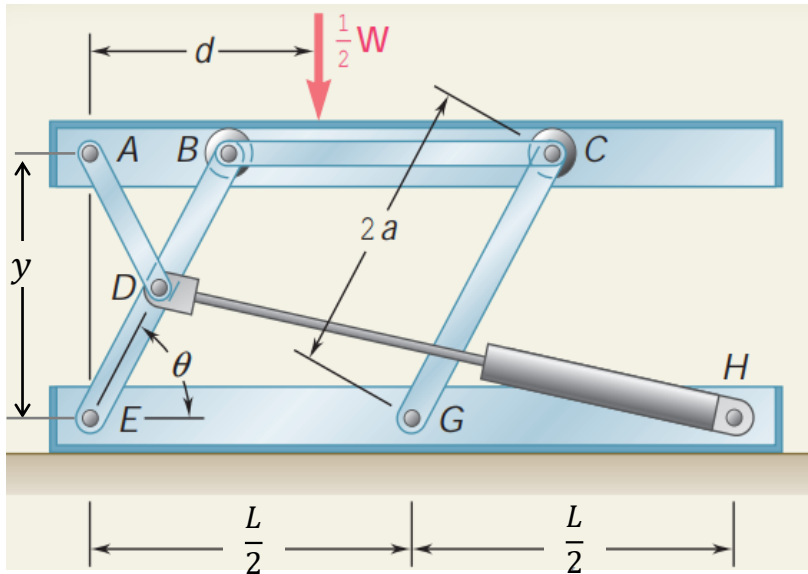
Consider the mechanism of half the hydraulic-lift table.



Suppose the virtual displacement is defined in such a way that conforms to how the assembly is being constrained to move.

Then only $\frac{1}{2}W$ and F_{DH} will give rise to virtual work done associated with the virtual displacement. Forces E_x , E_y and F_{CG} do not produce any work done simply because joints E and G are fixed.

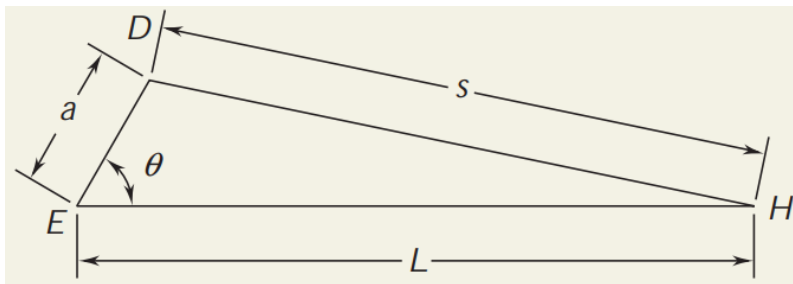
Example 5.3...



Based on the configuration of the system, we need to define the relations between the various moving parts w.r.t. θ .

For the platform ABC where $\frac{1}{2}W$ is applied,

$$y = 2a \sin \theta$$

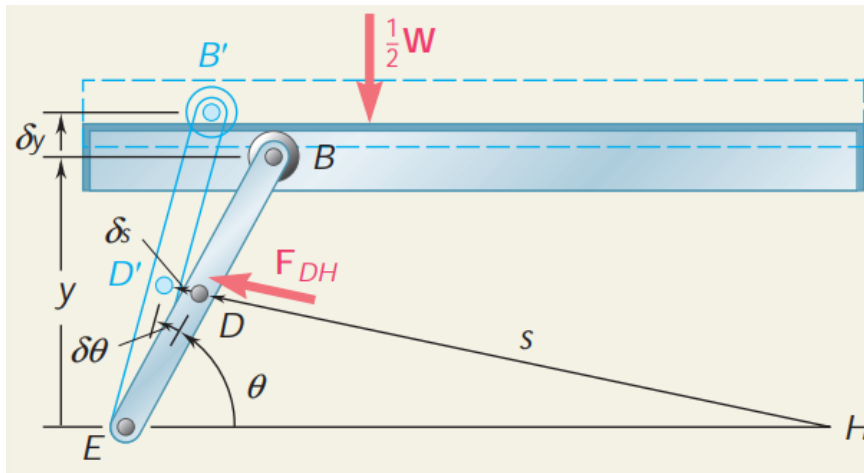


For point D , it is related to θ via the cosine law of $\triangle DEH$, i.e.

$$s^2 = a^2 + L^2 - 2aL \cos \theta$$

Example 5.3...

Suppose the system undergoes a small virtual angular displacement of $\delta\theta$ about E .



For $\frac{1}{2}W$, the virtual displacement of the platform is given by

$$\delta y = \frac{dy}{d\theta} \delta\theta$$

$$\Rightarrow \delta y = \frac{d}{d\theta} (2a \sin\theta) \delta\theta = 2a \cos\theta \delta\theta$$

For F_{DH} , the virtual displacement is

$$\begin{aligned} \delta s &= \frac{ds}{d\theta} \delta\theta \\ \Rightarrow \delta s &= \frac{aL \sin\theta}{s} \delta\theta \end{aligned} \quad \begin{aligned} s^2 &= a^2 + L^2 - 2aL \cos\theta \\ \Rightarrow \frac{d}{d\theta} (s^2) &= \frac{d}{d\theta} (a^2 + L^2 - 2aL \cos\theta) \\ \Rightarrow 2s \frac{ds}{d\theta} &= 2aL \sin\theta \Rightarrow \frac{ds}{d\theta} = \frac{aL \sin\theta}{s} \end{aligned}$$

Example 5.3...

Finally, according to the principle of virtual work,

$$-\frac{W}{2}\delta y + F_{DH}\delta s = 0 \Rightarrow -\frac{W}{2}2a\cos\theta\delta\theta + F_{DH}\frac{aL\sin\theta}{s}\delta\theta = 0$$
$$\Rightarrow F_{DH} = W\frac{s}{L}\cot\theta$$

Given the dimensions, $\theta = 60^\circ$, $a = 0.70 \text{ m}$, and $L = 3.20 \text{ m}$,

$$s^2 = (0.7)^2 + (3.2)^2 - 2(0.7)(3.2)\cos 60$$
$$\Rightarrow s = 2.91$$

With this, the piston force needed is

$$F_{DH} = (1000)(9.81)\frac{(2.914)}{(3.2)}\cot 60^\circ = \mathbf{5157 \text{ N}}$$